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Wildlife conservation and forestry: concerns and policy developments in India

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*As long as there is wildlife in the forest,
earth will provide shelter to man and his generations*
– old sanskrit saying (Karanth, 1985)

SUMMARY

India's geographical situation and diverse climatic features have endowed the country with a wide variety of biological communities. In the early years of forest conservation neither forest policy nor forest laws paid any regard to wildlife. Wildlife was 'game' and was plentiful. The food crisis of the second world war gave rise to the extension of agriculture at the expense of community forests. With independence, land distribution acquired an additional social hue – land for the landless – and even reserve forests were not spared. The extent of destruction would have been far worse if it had not been for the activities of some ardent friends of wildlife. Mrs. Indira Gandhi, a few articulate wildlife conservationists and the support of the forest department combined to provide the alchemy to reverse the situation. The paper discusses the principal laws introduced to conserve nature, laws which have often had beneficial effects for forests.

Keywords: forest reservation; India; wildlife protection.

INTRODUCTION

Over the centuries, people in India had a fascination and respect for nature. Ishopanishad, which dates back over 2,000 years, thus says 'All in this manifested world, consisting of moving and non-moving are protected by the Lord. Use its resources with restraint. Do not grab the property of others – distant and yet to come'. Emperor Ashoka's edicts (272-232 BC) proclaim the value of wildlife conservation and prohibit the burning of forests. In India, as in most traditional societies, people have lived for centuries without impinging on nature, with practices that ensured utilization of natural resources without degrading them (Anon, 1984).

The Directive Principles of State Policy of the Indian Constitution spell out the nation's commitment to protect the environment. Article 48-A of the Directive Principles states 'The state shall endeavour to protect and improve the environment and to safeguard the forests and wildlife of the country'. Environmental protection is also enshrined as a fundamental duty of the citizen. Article 51-A(g) mentions that it shall be the duty of every citizen of India 'to protect and improve the natural environment, including forests, lakes, rivers and wildlife and to have compassion for living creatures'.

Unfortunately, the situation today is the exact opposite of what one might expect from long-standing tradition and even the more recent pronouncements in the Constitution of India.

India's location in the tropics at the zone of confluence of three major bio-geographic realms viz. the Palearctic, the Ethiopian and the Indo-Malayan has endowed it with a great variety of biomes and biological communities. This bio-geographic advantage is further enriched by a variety of land forms, diverse geological and climatic attributes, a long coastline and many oceanic islands. The Trans-Himalayan cold desert, the scorching Thar desert, arid savannas, wet grasslands, steaming mangroves, dry thorn scrub and the temperate, deciduous and tropical rain forests are repositories of rich biodiversity. This is reflected in the hitherto recorded tally of 65,000 species of animals including 2000 species of fish, 140 amphibians, 420 of reptiles, 1200 of birds and 340 mammals (Panwar, 1993).

Unfortunately, much of this vast asset has been badly depleted during the last century. The cheetah, of which Akbar the Great (1556-1605) is said to have kept 3,000 with which to hunt antelope, is now extinct. The great Indian rhinoceros, which was hunted in the Indus Valley by the Moghuls as late as 1590, survives only in a few isolated Sanctuaries in India and Nepal. The population of Kashmir stag has been reduced to fewer than 250 individuals. The Asiatic lion once found throughout the open forests north of Narmada river now occurs only in Gir Forests of Gujarat, where fewer than a few hundred animals remain. Probably no more than 2,000 wild buffaloes exist in scattered forests. Of the 'herds of thousands' of barasinga seen by Pollock

and Thom (1900) in Assam and Bhutan, only some 4,000 survive in the country and in the world. Where Forsyth (1889) observed herds of blackbuck 'which must have numbered a thousand or more individuals', none are now seen and the species is in a critical stage of survival. The other large mammals – the India gazelle, nilgai, chital, gaur and so forth – which were exceedingly common have all declined drastically in number. The ubiquitous tiger has become scarce in most parts of the country (Schaller, 1967).

Wildlife cannot be conserved without saving the forest and the only forests left in the country are the reserved forests. More than half of these reserved forests are badly degraded.

RESERVATION OF FORESTS

The concept of reservation of forests developed out of the recommendations in the British Association for Advancement of Science's Report in 1851. Some of its main points were

- That over large portions of the Indian Empire, there is, at present, an almost uncontrolled destruction of forests.
- Wherever supervision has been exercised, considerable improvement has taken place.
- That special attention should be given to the preservation and maintenance of the forests occupying tracts unsuited for culture, whether by reason or altitude or peculiarities of physical structure.
- That in a country to which the maintenance of its water supplies is of such extreme importance, the indiscriminate clearance of forests around the localities whence those supplies are derived is greatly to be deprecated.

The subject was taken up by the Association because of alarming reports from India of large-scale burning of forests in several parts of the country for shifting cultivation. The concern was essentially the likely impact on soil and water in the tropics and neither aspects of biodiversity nor of wildlife had come in. Thus, the Indian Forest Act of 1865 as amended in 1878 and finally in 1927, which was in force over major parts of British India, or its modified versions because of local factors in other provinces, did not deal with wildlife at all. The only reference to wildlife in the Forest Act of 1927 is in the definition of forest produce which states 'wild animals and skins, tusks, horns, bones. . . and all other parts of wild animals . . .' constitute forest produce only if found in or brought from reserved forests. Thus, the Assam Legislature had to legislate specially during 1930s declaring rhino horn a forest product, wherever found, to facilitate prosecution of offenders (Agarwala, 1985).

A few laws were passed such as the Elephant Preservation Act, 1879 (to regulate shooting of female elephants) and the Wild Birds and Wild Animals (Protection) Act, 1912 which provided for closed hunting seasons and listing of animals whose hunting in reserved forests required a licence (Kothari *et al.*, 1989). It is thus not surprising that in the first National Forest Policy of India (1894) there is no

mention of wildlife. This is to be expected as at the time the concern of the government was mainly the extension of agriculture and hence the policy favoured the release of reserved forests except those in areas liable to erosion, if the demand for agriculture came up and could not be met from unreserved areas.

Between 1894 and the Second World War, wildlife was game for the hunters and poachers. In the native states, forests with wildlife and marshes with a bird population were the preserves of royalty for hunting. By and large, while in some cases it did result in heavy decreases in wildlife population (the population of Asiatic lion, found in only one forest, had fallen to some hundreds) the monopoly of the royal families had kept out the rest of the hunters. This in itself provided protection to wildlife. Some of the best Sanctuaries in India today were once the hunting reserves of the kings. As gun licence rules were rigid in other forests, wildlife remained reasonably safe.

In British India, in line with the practice of native kings, the Viceroy had his own reserved hunting reserve. Many British civil servants were keen hunters. In the reserved forests, rules provided for regulating shooting by the system of shooting licences, but there were none for promoting the cause of wildlife or for the control of hunting outside these forests. It required special legislation in 1936 to set up Hailey's National Park (since renamed Corbett National Park), the first of its kind in India.

During the war years of 1939 to 1945, many forests in India were over-exploited to supply timber for the war efforts and the military, with their jeeps and automatic weapons, inflicted considerable damage on wildlife in addition to 'normal' poaching which continued unabated (Gee, 1964). But the worst was yet to come.

During the later years of the war, a food crisis loomed large over the country and it was decided to promote the extension of agriculture. Under the 'Grow More Food Campaign', community privilege areas, which were set aside at the time of reservation to meet the rural needs of firewood, small timber and grazing and were approximately double the extent reserved, first fell to the axe. With Independence, the subject acquired a social dimension, that of land for the landless, and conversion of forested areas into agriculture received government support. The earlier requirement of a buffer zone between reserved forests and cultivation was given up and agriculture extended right up to the borders of reserved forests. The sanctity attached to the permanence of reserved forests was ignored. Development of irrigation and production of electricity were the slogans of the day and good forests in the valleys which were the locations of such projects were lost. At this time, concern for the forests was at its lowest ebb and evacuees were even resettled in cleared reserved forests lying within the catchments of the projects. Another sign of the times was that crop protection licences for use of guns were issued freely. For a decade after Independence, malaria was still unconquered and it held the human population in check in the heavy rainfall belts, but with its eradication thereafter no forest was safe from the influx of people and

their cattle. Wildlife was described as wild beasts, a menace to be subdued. Gone for ever the days of Gee when 'if one loses his way there is little chance of returning' (Gee, 1964). Surprisingly, even the National Forest Policy of 1952, which strongly urged conservation of forests, is silent on the subject of wildlife.

WILDLIFE PROTECTION: THE BEGINNING

According to Hailey, wildlife 'has few friends and many enemies'. Yet, if in India the situation of wildlife today is far better than in the 1950s, it is because of its few friends. While the forest and its conservation still had no supporters, apart from the dumb forestry service, in the case of wildlife, a few naturalists and many hunters, who were also lovers of wildlife, took up its cause. Dunbar Brander, F.W. Champion, Corbett, Jayachamaraja Wodeyar, Dharmakumar-asinhji, Salim Ali and Gee are the founder fathers of wildlife conservation in India. Chaturvedi, the first Indian Inspector General of Forests, availed himself of their support to suggest to the Government of India the setting up of the Central Board of Wildlife, later renamed the Indian Board for Wildlife (IBWL). This is a body with neither legal nor administrative powers, but due to the stature of its members, it has attained a status denied to most legally constituted authorities. During her Prime Ministership Indira Gandhi, a great lover of nature and wildlife, became the Chairman of the Board, and since then, while the forest portfolio would normally be the charge of a Minister in the Government of India, the Prime Minister continues to be the Chairman of IBWL. Political leadership committed to the subject, a few articulate wildlife conservationists and the khaki uniform of the forest department provided the alchemy for successful wildlife conservation efforts in India. This is where it has scored over the conservation of the forest, which has consistently lacked the support of articulate conservationists.

One of the first decisions of IBWL was that as wildlife in India existed in forests under the control of forest departments, its preservation should be the responsibility of the forest departments. Some of the other major recommendations of IBWL have been:

- (a) There should be unified comprehensive legislation for wildlife conservation in the country. The Wildlife (Protection) Act of 1972 was the result. It should be mentioned here that in 1972 'forest' was a State subject in the Constitution of India and it was well within the individual State's power to decline to go along with the model suggested by the Central Government. Yet the leadership of the late Indira Gandhi as Prime Minister assured the Act's acceptance by all the States (except Jammu and Kashmir, which followed it with certain amendments to suit its local requirements).
- (b) There were only 33 wildlife reserves in 1952. The Board suggested that considering the large number of habitats and the serious depletion of wildlife in the country, at least 4% of the land area of the country be set

apart for wildlife. Since then the number of Sanctuaries and National Parks in India has increased to 451 (Kothari *et al.*, 1989), covering some 4.35% of area of the country.

- (c) A model open Zoological Park be established in Delhi. As most zoos in India then had animals in cages, generally managed by municipal bodies, this recommendation was a milestone. The Delhi Zoological Park was the forerunner of a large number of open parks set up by the forest departments near other cities in the country.

On the recommendations of IBWL, State Wildlife Boards have been set up in most States and a Wildlife Institute of India has been formed. Incidentally, it is only since 1958 that wildlife management has been included in the training courses for forest officers and rangers.

A brief mention of the schizophrenic activities of foresters in the 1970s needs to be made. In its chapters on forestry the National Commission on Agriculture criticised the poor yield and economic returns from the natural forests and urged their conversion to plantations. Plantation programmes hence secured budgetary support of the Government of India to the State Forest Departments, otherwise starved of funds. The stature of a government department (and its prospects) largely depends on its budget and accordingly, large-scale reforestation programmes were taken up. This was also the period when wildlife conservation was gaining importance and the associated creation of Sanctuaries and National Parks (which, under the Forest Act, merely required government notification) went ahead. Thus arose the incongruous spectacle, quite often seen, of plantations in clear felled forests, within Sanctuaries and National Parks.

WILDLIFE PROTECTION: LEGISLATION AND FUNDING

The Wildlife (Protection) Act, 1972 was the first comprehensive law in India providing for protection of wildlife and its management. Being uniform, it has provided for coordination among States, a feature which was earlier missing. Its provisions deal with hunting of wild animals (and total protection to those on the endangered list); authorities to be appointed or constituted; formation of Sanctuaries, National Parks, Nature Reserves and closed areas; regulation of trade or commerce in wild animals, animal articles and trophies; and prevention and detection of offences (Saharia, 1982).

A unified organizational set-up for wildlife administration and management laid down in the Act has resulted in the formation of wildlife wings within the forest departments in almost all States. Wildlife is presently managed by the Chief Wildlife Warden of the State, a Forest Officer of the rank of a Conservator or above, under the Chief Conservator of Forests.

Funding of development programmes in the Government sector is through projects included in the country's Five Year Plans. What is outside the Plan has to be

supported by the State budget and both forestry and wildlife conservation have no special priority in this. Accordingly, forest departments opt for schemes included in the Plan and particularly those which secure budgetary support from the Central Government. The catch is that most of these projects stipulate, among other conditions, non-working or reduced working of the forest areas involved. The Chief Conservator of Forests has to impress on the State Government the financial assistance to be received from the Government at the Centre, while playing down the loss of revenue implied. With the Central Government, he has to emphasize the financial sacrifices to be made by the State in implementing the project. This tension between the attributes and wishes of Central Government officials and the attributes and capabilities of State officials has increasingly been resolved in terms of greater acceptance of central funding and hence lower revenue production from the forest. For the forester this has meant better protection with increased staff, equipment and vehicles in the project areas.

Sanctuaries and National Parks have brought the younger generation and the urban people closer to the forest, evoking a responsive chord in support of the forester. During nine years of tenure as the Head of the forest department in Karnataka, I also found that what could not be achieved through the bureaucratic maze was easily done when we had the Chief Minister with us, camping in a tranquil forest guest house, in a Sanctuary.

Project Tiger, conceived and implemented by the forester, has been acclaimed as one of the most successful projects in India. It was the brain child of the late Mr. Sankala, former Chief Conservator of Forests of Rajasthan. Demographic pressure both for extension of cultivation and use of wood as fuel coupled with heavy grazing has degraded large tracts of forests. Added to the shrinking depleted habitat, pressures of legal and illegal hunting had brought the status of wildlife in the country to a low ebb in the 1970s. As against the turn-of-the-century estimate of the Indian tiger population of between 20,000 and 40,000, the 1972 census estimated its numbers to be just over 1800. The project taken up in 1973 is designed to secure the long term survival of the tiger through large protected area units, caring for whole eco-systems and thereby the habitat and the prey and eventually the tiger. The seed money for the project was provided by World Wide Fund for Nature (Panwar, 1993).

One of the main lessons learnt from the Project Tiger is that with the density of population that exists within the forests in most parts of India, it is extremely difficult to achieve forest protection and wildlife conservation. It is the shifting of the human population from the Sanctuaries that has made the difference. Another equally important lesson learnt is that without restoration of degraded areas and improvement of the economic condition of the people, who presently depend on the forests for survival, forest and wildlife conservation is not possible.

The success of Project Tiger has resulted in taking up similar projects for the lion, the Kashmir stag, the Himalayan musk deer, the brow antlered deer and lately for the

elephant. The concept of habitat management, coupled with the restoration of surrounding areas, is also being extended to the National Parks and Sanctuaries, with financial support of the Central Government to the forest departments. This implies total closure to working in core areas, no grazing in National Parks and regulated grazing in Sanctuaries. Control over grazing is a measure otherwise impossible to enforce in the forests as it lacks the support of the State governments.

One of the most significant steps towards wildlife conservation in India was the framing of the National Wildlife Action Plan. At a meeting of the Indian Board for Wildlife in 1982, the then Prime Minister, Indira Gandhi, had outlined a broad framework for wildlife conservation. Based on this, the National Action Plan was formulated with the following objectives (Kothari *et al.*, 1989):

- Establishment of a representative network of protected areas
- Management of protected areas and habitat restoration
- Wildlife protection in multiple use areas
- Rehabilitation of endangered and threatened species
- Captive breeding programmes
- Wildlife education and interpretation
- Research and monitoring
- Domestic legislation and international Conventions
- National Conservation Strategy, and
- Collaboration with voluntary bodies and non-Government organizations.

The first three of these provide for considerable improvements, with financial support, to management of forests. A case of forests riding piggy-back on wildlife!

One of the measures adopted by the forest departments to reduce the antagonism of the local population was the provision to compensate for losses suffered by villagers due to wildlife. Initially this related to cattle kills due to carnivora in or around Sanctuaries and National Parks. It was found that as a measure of vengeance or to avoid further killings, the carcasses of the cattle were smeared with poison, which would result in the death of the concerned tiger or panther. Suitable compensation was a solace. This was next extended to the rest of the forests and thereafter to damage to agriculture as also to injury or death of human beings.

FOREST AND CONSERVATION POLICY TODAY

In 1976, the Indian Constitution was amended to shift forest from the State list to concurrent list, that is subjects comprehensively dealt with at both Central and State government levels but with Central Acts over-riding State Acts. This was feasible only because of the dynamism of the late Indira Gandhi as Prime Minister who had piloted the subject in the National Parliament. The first statute consequential on this change came in 1980 with the Forest Conservation Act, which restrained the powers of the State Governments to release forests or divert forests for alternate use, without

Government of India's approval. This one measure has reduced official diversion of forests which averaged 154,571 hectares per year between 1952 and 1980 to 14,351 hectares since then (Anon, 1989).

A policy statement of a department without legal support has no meaning. Thus, while the 1952 National Forest Policy prescribed principles for sound management, including that of private forests, through an Agrarian Reforms Act, private forests in one of the States were taken over for distribution, after clear felling, to the landless for agriculture. The National Forest Policy of 1988 can claim, for the first time in the history of forest management in India, official backing of the Central Government to implement it. It is not a matter of the calibre of the State and Central Government politicians but the fact that in a democracy, with elections once in five years, with decisions made in Delhi rather than in the States, where the subject takes on the hue of human beings versus forests, that makes the difference.

The latest National Forest Policy, which incidentally is the first forest policy to include conservation of wildlife, has as one of its basic objectives 'conserving the natural heritage of the country by preserving the remaining natural forests with the vast variety of flora and fauna, which represent the remarkable biological diversity and genetic resources of the country'. The effect of this policy decision will be explained with an example. In 1960s, the Sharavati hydro-electric Project had accounted for the clearance of 27,475 hectares of prime forests for submersion under water. The forest area cleared averaged 29.04 hectares per megawatt (MW) of electricity to be generated. The object of the forest department to the project was not only brushed aside, but rehabilitation of project evacuees was also taken up in enclaves in reserved forests. As against this, the proposed Aganashini hydro-electric project (in the same State, Karnataka) has not received the Government of India's clearance in spite of the loss of forest due to submersion averaging only 3.49 hectares per MW of electricity to be generated. The forest department's stand that the region has good natural forests and wildlife has been sustained (Shyam Sunder, 1994). It should, however, be added that the Policy dictate has succeeded only because of the strong support provided to it by the Forest Conservation Act.

Under the heading 'Essentials of forest management', the Policy stresses the need to strengthen and extend the network of National Parks, Sanctuaries, Biosphere reserves and other protected areas for the conservation of the total biological diversity. It has also stressed that 'forest management should take special care of the needs of wildlife conservation and forest management plans should include prescriptions for this purpose ...'. These policy aims are being implemented through a system of incentives, Government of India offering up to 100% financial assistance for such programmes taken up in the States by forest departments.

ENVIRONMENTAL POLICY AND THE FUTURE

The National Action Plan proposed by the IBWL referred to earlier had suggested the formulation of a National Conservation strategy for the country.

The National Strategy and Policy Statement on Environment and Development, Government of India, 1992 is the result. This policy statement has a much wider scope than the National Forest Policy and can take support from a large number of enactments. It has identified the causes of ecological degradation in the country, of the deteriorated condition of forests and the depleted status of wildlife. The major reasons lie outside the forestry sector, but impinge on it. It has strongly recommended measures to be taken for the conservation of forests and wildlife both through direct action in these sectors and supportive action in countering or reducing the damaging influences.

On the policy front, the future of forests and wildlife in India is far brighter than it was a decade ago. Yet there remains a difficulty. During the war years, army personnel with their jeeps had played havoc with wildlife. Since then jeeps have played havoc with the forests. Many of the younger generation of foresters in India have lost the art of walking through the forests. It is now inspection of plantations, of sites in which investment is made. With nil or little investment in the natural forests, no inspection is called for. This is a tragedy of the highest order for, without a feel of the natural forests and wildlife, there will be no genuine concern on the part of its guardians.

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